DOMAIN GUIDE

Base Observation And Fallback Review

This domain makes evidence from one chain safe to use elsewhere in a Bitcoin-backed USDC loan. The Base contracts settle the USDC; Bitcoin independently controls the collateral. A transaction appearing in a wallet, an updated application screen, or a missing lender response cannot bridge those authorities. This domain supplies trustworthy observations of what happened on Base and defines a narrowly limited fallback that may help return Bitcoin after an unclaimed repayment has been refunded.

> This is the domain guide for Base Observation And Fallback Review, one of the five domains of Bitcoin-backed USDC one-loan system. The architecture guide, a separate document, explains how the domains work together.

It contains two distinct responsibilities. Base observation, finality, and reconciliation follows the approved deployment, records its events and their finality, and repairs the application's view when evidence changes. The evidence-conditioned validation-service fallback can publish a loan's precommitted secret after independently verifying one exact refund event. Observation reports facts; the fallback controls that secret's disclosure. Neither responsibility decides who deserves the collateral or signs a Bitcoin transaction.

Establishing a fact the rest of the system can use

The observer reads named contract events for funding, acceptance, repayment, claim and refund. It checks which chain, contract deployment, loan, transaction and block produced each event. An event first seen in a Base block is provisional. Even the intermediate “safe” status is only progress information: application-side lifecycle advances and Bitcoin handoffs wait until the block reaches **Base L1 (layer 1) finality**, the stronger settlement condition defined in the finality decision. That decision accepts additional waiting in return for one consistent rule at consequential cross-chain boundaries; it is not a promise of an exact waiting time.

The observer sends those facts to the application through a versioned event-ingestion handoff. The application records the source evidence and derives the status a participant sees. It acknowledges storage before the observer advances its durable cursor—the saved position from which it can safely resume. If an acknowledgement is lost, the observer sends the same event again; the stable identity prevents a duplicate. This event history was selected so that a restart or provider change can reconstruct what happened, instead of trusting a mutable loan snapshot or an independently edited watcher status.

Alchemy is the primary Base read source and QuickNode is the fallback. One healthy configured provider is sufficient for a finalized Base fact; the second provider supplies failover and disagreement detection. If both are unavailable or disagree, existing finalized projections remain visible while new consequential actions that depend on Base are blocked. When a chain reorganization displaces an observed event, the observer retains it as noncanonical evidence,

appends a correction, and replays from the last common finalized block. An operator may request a recheck through the application, but cannot change the fact, override the result or advance the observer's cursor.

The responsibility also reconciles independently sourced Bitcoin evidence. Bitcoin lifecycle gates require fresh agreement between the configured observers, unlike the one-healthy-source Base policy. Conflicts and outages block dependent automation; repeated reports update the same incident, recovery replays from the last agreed state, and a failure after borrower acceptance receives immediate critical treatment. Vault payment transfers have another limited meaning: exact transaction and nonce evidence can update a payment's own record, but only the named finalized contract event can establish loan funding or repayment. Even a confirmed standalone participant transfer does not advance the loan. The observer's precise rules retain the provider, retry, escalation and payment-profile distinctions.

When the borrower repays but the lender does not claim

A concrete case explains why the fallback is separate. Assume exact repayment has finalized by the repayment deadline, **D**. Normally the lender claims the USDC by revealing a secret called **Y** before the claim cutoff. Public Y then lets the Bitcoin finalizer complete a recovery transaction that the borrower already signed for a fixed destination. If the lender cannot or does not reveal Y, valid repayment alone does not release the Bitcoin.

At **C = D + 172,800 seconds**, two days after the repayment deadline, Base permanently disables the lender claim and freezes the current borrower refund recipient. Anyone may call the fixed-effect refund; the caller cannot choose where the money goes. A borrower concerned about access to the original recipient may have switched it to the committed borrower protocol account before C; the choice is frozen at C. Once the refund's exact **UnclaimedRepaymentRefunded** event reaches Base L1 finality, the observer can report that the exact USDC repayment was returned once. The unclaimed-repayment flow makes claim and refund mutually exclusive on Base. The refund does not require the fallback service, and the Bitcoin output remains separately unspent.

The selected fallback is an **evidence-conditioned validation service**, or **ECVS**: one contracted independent service that holds a unique random secret **S** for each eligible loan. Before that loan is constructed and signed, it returns only **H(S)**, a hash commitment used later to check the disclosed secret. The Bitcoin recovery script must already contain this commitment. The company, borrower and lender do not receive unrevealed S and cannot command an early release.

The ECVS does not accept the application observer's conclusion as authority. It independently checks the configured Base deployment, loan, exact refund event, amount, frozen recipient, provenance and L1 finality. Only then may it publish the matching S durably and publicly. Bitcoin finalization separately checks the publication, commitment, existing signatures and still-unspent escrow, then uses S in the **same borrower-signed recovery transaction** that could have used Y. No fresh borrower signature, second fallback transaction family, service signing key

or new destination is introduced. Publication, broadcast and confirmed Bitcoin recovery are separate steps; the conditional recovery flow and fixed Bitcoin handoff preserve those boundaries.

Limits and remaining recovery paths

This model is selected, but production use remains gated by counsel approval and implementation evidence. Until that gate is satisfied, a new loan must not commit H(S) or advertise this fallback. An existing Y-only escrow cannot acquire the capability later. The selected design avoids giving the service a Bitcoin key or discretionary evidence-arbitration role; its secret can still be disclosed too early, withheld, lost or compromised. Those risks are why independent controls and the activation gate remain material.

Missing Y, elapsed time, an application assertion, a raw token transfer or the earlier **Repaid** event cannot trigger S publication. If the service is unavailable, evidence is wrong, or S cannot be published, the Base refund remains complete and Bitcoin recovery through S remains unavailable. The lender's pre-signed Bitcoin route can still mature at D plus three days under Bitcoin's consensus time rules. A separate borrower long-stop can mature at D plus ten days without Y or S, but does not cancel the already-mature lender route; one confirmed spend wins. Whether a loan without a production-activated S commitment has that long-stop is an open question: a build constraint requires every newly constructed escrow to include it and the long-stop recovery flow defines it, while the missing-Y security boundary says such a loan has no borrower fallback Bitcoin route and the unclaimed-repayment flow lists the remaining routes without it; the records do not settle whether these describe different loans or contradict each other. Mutual cooperation is another separately authorized route. These are residual Bitcoin risks, not evidence of borrower default after valid repayment or a guarantee that collateral will return.

Automatic organization broadcast of lender-default packages and institutional collateral recovery remain outside the selected operating boundary. Monitoring, package storage and the ECVS give the company no collateral key, sale authority or discretionary payout power. Exact contracts and the two responsibility records linked above remain the owners of every rule summarized here.

The architecture guide shows where this domain fits in the system, and the domain record lists exactly what the domain owns and excludes.

Other guides

This guide refers to these separate documents of the System Guide:

- How the Bitcoin-backed USDC one-loan system works (architecture guide)

Technical records

The exact rules, contracts and conditions live in these records of the design package:

- **Domain record:** Base Observation And Fallback Review.
- **Responsibility records:** Base Observation, Finality, And Reconciliation; Evidence-Conditioned Validation-Service Fallback.
- **Flow records:** Borrower Long-Stop Recovery; Conditional ECVS Borrower Recovery; Valid Repayment Without Lender Claim.
- **Interface contracts:** BaseEventObservationV1 And Correction; Finalized X, Y, Or S To Fixed Bitcoin Action.
- **Invariant:** Missing Y Residual Lender Risk.
- **Security boundary:** Valid repayment plus missing `Y` leaves Bitcoin unsettled.
- **Scope boundaries:** Automatic Lender-Default Broadcast; Institutional Last-Resort Collateral Recovery.
- **Selection record:** Base L1 finality and reorg reconciliation.